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Response of Mother and Infant Squirrel Monkeys to Separation and Disturbance

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VOGT, J. L. AND S. LEVINE. *Response of mother and infant squirrel monkeys to separation and disturbance.* PHYSIOL. BEHAV. 24(5)829–832, 1980.—Plasma cortisol values were obtained from squirrel monkey mothers and infants after two types of brief separation: infant removed to a nearby cage and infant removed to a different room. Cortisol values were also obtained during basal conditions, and after 30 min exposure to an unfamiliar juvenile conspecific when mothers and infants were allowed to maintain contact. For either separation condition, both mothers and infants exhibited significant cortisol elevations, and overall, infants' corticoid values were much greater than mothers'. Behavioral measures were also obtained from mothers, who during both separations tended to show marked behavioral agitation. The partner's presence or absence during separation did not produce differential behavioral or corticoid responses by either mother or infant. However, the presence of the unfamiliar conspecific induced clearly significant cortisol elevations in the infant but only slight elevations in the mother.

Mother-infant Plasma cortisol Separation Squirrel monkey

PRIMATE infants are quite sensitive to disruption of the mother-infant bond, and one of the criteria of infant attachment has been the pattern of behavioral changes in response to separation [1, 2, 3, 7, 10]. There are also physiological consequences of mother-infant separation. Reite and colleagues have demonstrated with pigtail macaques the deleterious effects of long-term maternal separation upon infant heart rate, body temperature, and REM sleep all of which show marked decreases from baseline on the first night of separation and thereafter [16, 17, 18]. In our laboratory we have used the pituitary-adrenal system to assess the effects of brief separation and have examined mother as well as infant squirrel monkeys; both members of the dyad show dramatically elevated plasma cortisol levels after a brief, 30-min separation [13,14]. These results were obtained whether mother or infant was removed from the home environment and even if the infant was "aunted" (retrieved and carried by another female) when left in the group [5]. Thus, corticoid responsiveness has been shown for squirrel monkeys to be a highly sensitive measure of the arousal induced by separation and may reflect aspects of the state of the animal not readily apparent in the behavior. In this study we have examined further this pituitary-adrenal sensitivity by assessing for squirrel monkey mothers and infants the responsiveness to a brief separation when the other member of the dyad was present or absent.

There is behavioral evidence to indicate that separation under different conditions does significantly influence both members of the mother-infant dyad (see review, [15]). For 6-month old rhesus infants, a separation procedure allowing physical contact with the mother produces minimal disturbance [23], while a separation procedure involving visual and auditory but not physical interaction with the mother is more disturbing than is a separation procedure involving removal of the mother to a different room [19,20]. In previous squirrel monkey separation studies this variable has not been systematically manipulated. A total separation procedure where the mother or infant is removed to a separate room has been used by our laboratory (e.g., [14]) and by Jones and Clark [8] although they removed simultaneously all mothers and left all infants together in the home cage; and an adjacent separation procedure involving the removal of the mother to an adjacent cage was utilized by Kaplan [9]. All of these studies produced rather marked immediate changes in the behavior of the infant [8], of both mother and infant [9], or in the plasma cortisol levels of both mother and infant [14]. However, systematic comparisons of the effects of squirrel monkey mother-infant separation under different social environmental conditions have not yet been done. Therefore, in this study we examined both the hormonal and behavioral responses of squirrel monkey mother-infant pairs to conditions in which either the separated individuals were completely isolated from each other or in which they were permitted visual and auditory but not physical interaction.

METHOD

Animals and Housing Conditions

The animals in this study were squirrel monkeys (*Saimiri sciureus*) of the Guyanese variety. The mothers had been imported as adults and had lived in our laboratory for several years. Three weeks before the start of testing four mothers and their 4-month old infants were formed into a group in a $1.83 \times 1.83 \times 1.22$ m wire mesh cage, with three sets of perches extending the length of the cage. Located in an open air shed with an opaque roof, the cage was subject to natural climatic conditions, and heat lamps and space heaters provided addi-

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tional warmth. Standard monkey chow, orange drink, and water were continuously available, and fresh fruit was provided twice weekly as a dietary supplement. At least 60 min before each test session a 51×46×46 cm wire mesh stimulus presentation cage was placed adjacent to the group cage.

Procedure

The details of the four test conditions were as follows: (1) *Adjacent Separation:* The experimenters entered the cage shed, one experimenter removed the specified infant, left the mother with the remaining three mother-infant dyads in the group cage, and placed the separated infant in the stimulus cage next to the group cage. Following the 30-min separation, blood samples were taken from the separated mother and infant. (2) *Total Separation:* The same procedure was used as in Adjacent Separation except that the separated infant was placed in another small cage, carried to a separate building, and therefore totally isolated from the group. Blood samples were again taken from the separated mother and infant. (3) *Stranger:* As a control for the disturbance of infant separation, the experimenter briefly entered and left the mother-infant cage just before an unfamiliar 18-month-old male conspecific was brought into the cage shed. This juvenile was carried in a small cage and placed next the the group cage for 30 min, after which time blood samples were taken from all mothers and all infants. (4) *Basal:* The experimenters entered the shed area and rapidly obtained blood samples from all mothers and all infants in order to determine resting cortisol levels.

During the study, there were 12 test sessions, conducted at weekly intervals between 1230 and 1430. Each of the four pairs was subjected to the two separation conditions, Adjacent Separation and Total Separation, with at least 3 weeks between these two conditions for a particular pair. Any test session imposing a separation condition involved the separation of only one mother-infant pair. All animals were tested in the Basal and the Stranger conditions twice, once each in the first 4, and once in the last 4, weeks of testing. The Basal condition was performed early and late in the study period so as to examine the possibility of chronic elevations in cortisol levels across the study, as found by Coe *et al.* [4].

Behavioral responses of the mother but not the infant to separation were obtained. Infants were not observed as the infant separation procedure was always confounded with placement of the infant in a novel environment. For the purposes of comparison a set of undisturbed baseline observations on the mother were also gathered. These baseline observations focused on the movement of the mother and were obtained on one afternoon early in the study period. For both the separation and baseline sessions, the observer was behind one-way glass during the 30-min tests. These observation periods were divided into 5-min segments, with ten 30-sec intervals signaled by a click from an electric timer. Three behavior patterns, *Proximity, Movement,* and *Arm Reach,* were scored during alternate 5-min periods for a total of 15 min of data. The behavior patterns were defined as follows: *Proximity:* Location within 30 cm of the stimulus cage; *Movement:* Change of location of at least 15 cm and two successive movements interrupted by a stop in activity for 3 sec; *Arm Reach:* Reach through the wire mesh toward the stimulus cage. Proximity was scored only on-the-signal while the other two behaviors were scored by frequency of occurrence.

Plasma Cortisol Determinations

Animals were anesthetized with ethyl ether and 0.7 ml of blood was collected in heparinized syringes via cardiac puncture. Blood samples for each animal were obtained within 3 min of initial exposure to ether, and all samples were taken within 15 min of the start of the blood draw procedure. The blood was then centrifuged, and the plasma collected and frozen until assay. Cortisol was assayed using antiserum no. F2153 from Endocrine Sciences, Tarzana, California, as described by Klemm and Gupta [12]. Further details of blood sampling and radioimmunoassay for cortisol have been presented in previous publications [5,14].

RESULTS

For the infants, the presence of the mother had no effect upon the infants' plasma cortisol response to separation. Paired t-tests showed that Adjacent (mean=279.7 μg/100 ml) and Total (mean=248.1 μg/100 ml) Separation values did not differ ($t=1.40$). Further, there was no difference between the infants' first and second Stranger ($t=0.08$) or first and second Basal ($t=1.58$) samples. Since the two Separation and two Stranger values did not differ, these scores were averaged to yield mean Separation and mean Stranger scores. The two Basal scores were also averaged for a mean Basal value, which was used to reflect baseline corticoid activity. A repeated measures analysis of variance of each infant's mean value during each of the three conditions revealed a significant main effect for conditions, $F(2,6)=18.83$, $p<0.005$, and post hoc comparisons using Duncan's multiple range test [11] indicated that both the Separation and the Stranger values were greater than Basal (both $p<0.01$) (Fig. 1).

For the mothers, the presence or absence of the infant likewise had no effect upon the plasma cortisol response to separation. Paired t-tests for Adjacent (mean=168.9 μg/100 ml) and Total (mean=144.4 μg/100 ml) Separation indicated that these conditions did not differ ($t=0.83$). The first and second Stranger values also did not differ ($t=2.79$), but there was a slight but systematic rise from the first (mean=86.9 μg/100 ml) to the second (mean=110.9 μg/100 ml) Basal sample ($t=3.21$, $p<0.05$). Since the two Separation and two Stranger values did not differ, these scores were averaged to produce mean Separation and mean Stranger values. A repeated measures analysis of variance was then performed on the mothers' values during the four conditions (Basal 1, Basal 2, Stranger, Separation). A significant main effect for conditions was obtained, $F(3,9)=5.24$, $p<0.05$. Post hoc Duncan multiple range tests revealed that Separation was greater than either Basal 1 ($p<0.01$) or Basal 2 ($p<0.05$) and that Stranger was greater than Basal 1 ($p<0.05$) (Fig. 1). An overall comparison of mothers and infants used mean values at each condition and indicated that infants' plasma cortisol values (mean=206.5 μg/100 ml) were much greater than those of mothers (mean=128.8 μg/100 ml), $F(1,6)=23.65$, $p<0.005$.

The behavioral response of the mothers to separation was reflected primarily in increased activity which was directed toward the separated infant when it was present. The individual scores as well as the means for the three behaviors are plotted in Fig. 2. Because of the greatly divergent between-cell variances of these data, nonparametric (Wilcoxon signed-ranks) tests were used for statistical analysis [21]. Mothers had higher movement scores during both separation conditions than they did during baseline observations. However, because of one very low value during Total Separation,

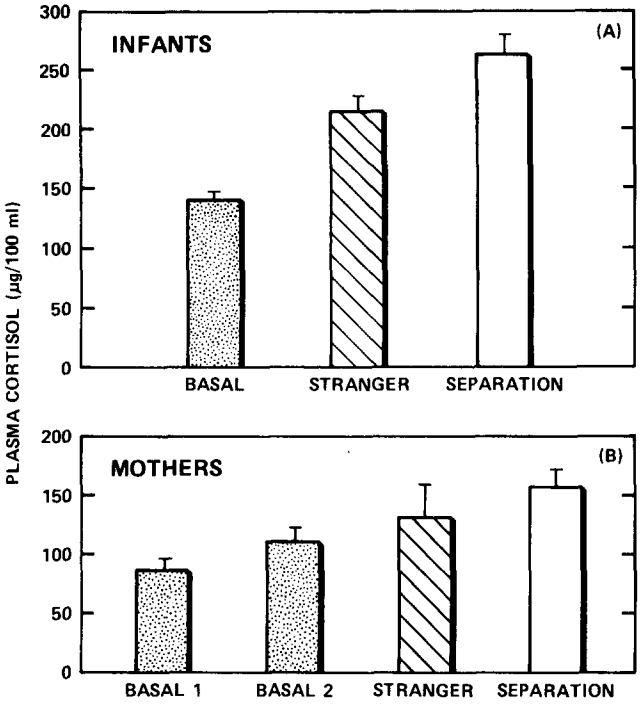


FIG. 1. Mean (± SE) plasma cortisol values of infants (A) and of mothers (B) for the experimental conditions.

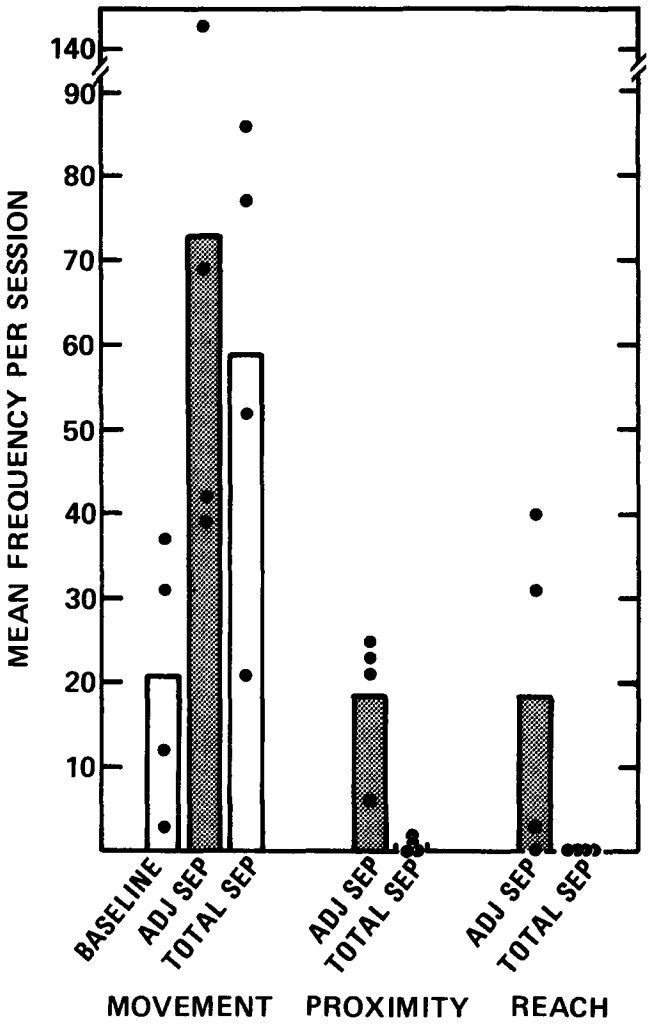


FIG. 2. Individual and mean frequency per session of *Movement*, *Proximity*, or *Reach* for each separation condition. Only *Movement* was assessed during undisturbed baseline conditions.

only the Adjacent Separation-Baseline comparison achieved significance ($p<0.06$, the lowest p possible with n=4). Proximity and Reach scores were greater during Adjacent than during Total Separation, although significantly so only for Proximity ($p<0.06$).

DISCUSSION

This study on squirrel monkey mother-infant separation found that Adjacent or Total Separation had comparable effects on the pituitary-adrenal activity of both mother and infant and on the behavior of the mother. While seemingly in contrast to studies in which rhesus infants show a far more agitated reaction to an Adjacent, as opposed to a Total, Separation [19,20], there are several major differences between the present study and the previous research. We used a brief (30-min) separation rather than a separation of several weeks, we tested a different primate species, the squirrel monkey, and we monitored the behavior of the separated mother rather than the separated infant. Also, our primary measure of the separation response for both mother and infant was adrenocorticoid activity. But our results do indicate that for brief separation in squirrel monkeys the proximity of the dyad during separation does not exacerbate the response and, conversely, total isolation of the mother and infant does not ameliorate the response.

Both Adjacent and Total Separation conditions had physiological and behavioral effects. The pituitary-adrenal responsiveness of both mother and infant to separation replicates earlier work [5, 13, 14] and extends the effect to an Adjacent Separation procedure. Furthermore, the present study indicated as did two others [5,13] that infants showed much greater plasma cortisol levels than did mothers. This experiment also assessed the mother's behavioral response to separation from her infant under different environmental conditions. The mother's behavior during separation has been examined in only one other study [9], where the mother's removal to an adjacent cage produced immediate behavioral disturbance. In the present study we found that both Adjacent and Total Separation seemed to produce marked behavioral agitation in the mother. Thus, brief separation seems so distressful for the squirrel monkey mother that the presence of her infant does not intensify her behavioral agitation.

While either separation procedure induced unambiguous behavioral and physiological responses by the mothers, the Stranger condition may or may not have brought about significant corticoid elevations in the mothers, depending upon the basal sample used for comparison. In contrast, the infants clearly exhibited elevated cortisol levels in response to the disturbance of the stranger. This significant elevation by the infants may reflect disturbance or fear of the unfamiliar conspecific, but it may also reflect heightened arousal or curiosity toward the novel social situation. Since the infants, who maintained contact with the mothers throughout this condition, did not approach or explore the new animal, it seems more likely

that the increased pituitary-adrenal output is indicative of fear. Regardless, the elevation to the stranger for the infants and possibly mothers was somewhat surprising in another context: the higher adrenocorticoid values occurred despite the physical contact between the dyad. In a previous study when a mother-infant pair was captured, pulled apart, but then immediately reunited, the physical contact during the 30 min after reunion seemed to inhibit pituitary-adrenal responsiveness in both members of the dyad [14].

If in fact mothers and infants respond with differential patterns of cortisol elevations to the unfamiliar animal, our results do not indicate whether these different patterns reflect response characteristics of mothers and infants or whether the time course of the pituitary-adrenal system differs for mother and infant squirrel monkeys, as is the case of adult and weanling rodents [6]. There is some evidence from squirrel monkeys for the time course hypothesis in that adrenocorticoid levels of infants remain elevated for longer periods of time than do levels of mothers [13].

Finally, a most striking aspect of these data is that the plasma cortisol levels of the infants were much greater than those of the mothers. This age difference has been reported in two previous squirrel monkey mother-infant separation studies [5,13], although not of the magnitude of the present study where infants' values were nearly twice as great as mothers'. Other primate species apparently do not show these differences in adrenocorticoid values between mothers and infants. Under basal and separation conditions rhesus mothers and infants exhibit comparable levels of plasma cortisol [22], as do human mothers and infants during undisturbed conditions (Kasting, unpublished observations). The functional significance, if any, of the squirrel monkey infants' higher cortisol values is not known at this time, and virtually no studies on the development of the primate pituitary-adrenal system have yet been conducted.

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